SELECTED ROBOTICS AI & SYSTEMS INTEGRATION PORTFOLIO

Brandon Gerbasi

> *"The part I enjoy most is final integration - taking AI, software, sensors, compute, electronics, and hardware and bringing the complete machine to life."*

Computer Science B.S. expected 2026 | Mathematics B.S. in progress | Relocating to Austin, TX

AI & Learning Qwen3 local LLMs, PyTorch, speech AI, wake-word training, RL/IL foundations	**Robotics Software** C++17, Python, ROS2, state estimation, telemetry, safety logic	**Physical Integration** Sensors, Jetson/NVIDIA GPUs, UAV hardware, CAD, wiring, fabrication
Systems & Deployment Linux, CUDA, systemd, authenticated APIs, multi-node health, testing	**Field Experience** Tesla manufacturing, customer-facing field work, travel, rapid cross-training	**Current Direction** Physical AI, robotics systems integration, edge autonomy, sim-to-real

2 Linux GPU nodes	**30B** local MoE brain	**32K** validated context	**19** registered tools	**15** automated tests

Palo Cedro, CA | Relocating to Austin, TX | 530-355-1711 | brandon.gerbasi@gmail.com | linkedin.com/in/brandon-gerbasi | github.com/gerbasi117

PROJECT 1 - OTTO: DISTRIBUTED LOCAL AI & ROBOTICS ASSISTANT

Purpose: Build a private, offline-first assistant that can hear, reason, use tools, coordinate across compute nodes, and safely interact with computers and future robotic systems without cloud inference.

AUDIO / WAKE CORPUS	SPEECH NODE	AI CORE	SAFETY & TOOLS	OUTPUT
Blue microphone 585 local WAVs 5,000+ mobile queued	faster-whisper medium RTX 3060 / CUDA authenticated STT API	Qwen3-30B-A3B RTX 3090 / 32K Q8_0 KV + flash attention	19 registered tools confirmation gates memory + compaction	Piper male TTS reports / system actions soundbar keepalive

VERIFIED BUILD STATUS

2 GPU nodes	32,768	152-164 tok/s	2.7-3.0 GB	PASS
Main + speech worker deployed	validated context / Q8_0 KV	Qwen3 benchmark range	measured VRAM headroom	15 tests + full validator

WHAT I BUILT

- Deployed a two-node local architecture with Qwen3-30B-A3B on RTX 3090 Main and faster-whisper medium on RTX 3060 Node 2.
- Built an authenticated raw-WAV STT service with **/health**, request limits, CUDA inference serialization, and systemd autostart.
- Refactored the original monolith into modular LLM, STT, TTS, memory, prompt, tool, system, and operations services with a 19-tool registry.
- Implemented protected confirmations, remote service restart, verified Node 2 reboot/recovery, bounded tool loops, atomic state, and history compaction.
- Optimized Qwen3 with the embedded chat template, full GPU offload, Q8_0 KV cache, flash attention, and automatic 14B fallback profiles.
- Loaded Piper once, chunked long speech, and used a persistent audio keepalive to eliminate clipped sentence beginnings and endings.

NEXT ENGINEERING MILESTONES

Strict Offline Lockdown Local model paths, public-network tool controls, and WAN-independent validation.	**Local Knowledge & Memory** Approved-folder indexing, structured memory, tasks, and project retrieval.
Production Wake Word Import 5,000+ mobile WAVs, deduplicate, normalize, train, and measure false activations.	**Custom Voice & Robotics** Train Otto's voice and connect real drone/smart-home protocols behind safety interlocks.

PROJECT 2 - CUSTOM FIELD ROBOTICS UAV PLATFORM

Purpose: Design and integrate a custom aerial robotics platform combining propulsion, flight control, telemetry, sensing, edge compute, safety logic, and staged autonomy/vision capabilities.

Mechanical Platform	Custom CAD and 3D-printed body/wings, carbon-fiber reinforcement, side-by-side battery bay, payload and balance planning
Power & Propulsion	Dual CNHL 9500mAh 6S architecture, Tekko32 65A ESC, 2807 1300KV motors, XT90 anti-spark, power-module and protection planning
Flight & Communications	Flight Core V2, u-blox M10 GPS, ELRS 915 MHz control, 915 MHz telemetry, mission/log workflow
Perception Sensors	RP-LiDAR S2, TFmini Plus forward/downward ranging, FLIR Lepton thermal camera, GoPro/visual payloads
Edge Compute	NVIDIA Jetson Orin Nano for ROS2, OpenCV, tracking, obstacle avoidance, sensor fusion, and autonomy work
Ground / Operator Tools	Python telemetry ground station, warnings, reports, logs, state estimation, and repeatable mixed-fault tests

6S x2	**2807**	**Jetson**	**LiDAR**
battery architecture	1300KV motors	Orin Nano edge AI	thermal + range sensing

INTEGRATION STATUS

- Completed initial manual flight/propulsion validation; the custom airframe and wings are being revised and reprinted for repeatable structured testing.
- Core flight-control, power, GPS, control-link, and telemetry architecture is integrated; Jetson and perception hardware are staged for activation.
- Next phase: stable-airframe flight logs, telemetry review, then staged ROS2, computer vision, object tracking, avoidance, and autonomous behaviors.
- The project is intentionally end-to-end: design, fabrication, wiring, software, sensing, testing, failure analysis, and operator tools belong to one system.

WHY THIS PROJECT MATTERS FOR INTEGRATION ROLES

It demonstrates the work I want to own professionally: integrating sensors, compute, control software, electronics, communications, mechanical hardware, and safety into a machine that must work outside a perfect lab environment.

SUPPORTING ROBOTICS & AI PROJECTS

C++ Robot State Estimator	**C++17, CMake, GTest**	Tracks position, heading, and velocity; handles GPS loss; scores confidence; produces repeatable CSV/Markdown/JSON reports and fault-test artifacts.
UAV Telemetry Ground Station	**Python, tests, log analysis**	Simulates/replays flight states, detects unsafe or high-current conditions, and generates operator-facing reports across normal and mixed-fault scenarios.
ROS2 Perception & Safety Demo	**ROS2 Jazzy, C++**	Separates range sensing, IMU input, safety control, and logging; demonstrates CLEAR, SLOW_DOWN, STOP_OBSTACLE, and STOP_TILT states.
Vision-Based Object Tracker	**Python, OpenCV**	Demonstrates real-time visual tracking and establishes the perception foundation for future UAV/robot autonomy.
Edge-Deploy Safety Monitor	**Python, automated tests**	Evaluates sensor/system conditions and produces explicit safe-state outputs for edge deployment.

VERIFIED SYSTEMS EVIDENCE

19	**15**	**2**	**32K**	**~153-164**
registered Otto tools	automated tests	Linux GPU nodes	validated context	tokens/sec benchmark

PROFESSIONAL EVIDENCE

- **Tesla Gigafactory Nevada:** four performance shout-outs in the first month; selected for overtime in the first week; cross-trained across multiple production areas earlier than standard.
- **Botco embodied-AI project:** wore sensing equipment while performing repeatable physical tasks to produce human-demonstration training data.
- **Six years of independent field/customer work:** route planning, deadlines, difficult interactions, documentation, and professional work without close supervision.
- **Hands-on technical background:** electronics, soldering, vehicle repair, heavy machinery operation/maintenance, CAD, fabrication, and Linux/GPU systems.

WHAT I BRING TO A ROBOTICS INTEGRATION TEAM

Systems View Trace failures across software, sensors, power, communication, compute, audio, and mechanics instead of treating one layer in isolation.	**Rapid Learning** Learn unfamiliar systems by building, testing, documenting, and iterating until the cause is understood - not just the workaround.	**Physical Ownership** Comfortable wiring, mounting, configuring, programming, testing, and repairing complete machines.
AI + Robotics Connect model training and AI software to ROS2, edge compute, sensor data, safety logic, and real hardware.	**Field Readiness** Comfortable with travel, customer sites, long deployment days, changing conditions, and direct responsibility for outcomes.	**Career Direction** Become exceptional at bringing AI-powered physical systems from separate components to reliable operation.

Available to relocate to Austin, TX | Available for field travel | github.com/gerbasi117